

Relationship Between Total Child Deaths and Pneumonia Deaths in Children, 2020

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Objective

This report examines the relationship between two 2020 country-level child mortality indicators: (1) the number of total child deaths and (2) the number of deaths due to pneumonia in children aged 1-59 months.

Data overview and preparation

Both datasets were reviewed first to understand their structure. Each dataset is in wide format, with country identifiers in the geo and name columns and years stored as separate columns. The analysis subsetted each indicator to the year 2020, then confirmed that the mapping inputs had one row per country before producing maps.

Indicator	Original dataset size	Country-code check	2020 values
Total child deaths	194 rows x 225 columns	0 duplicate geo rows	193 non-missing 2020 values
Pneumonia deaths, children 1-59 months	192 rows x 24 columns	0 duplicate geo rows	187 non-missing 2020 values

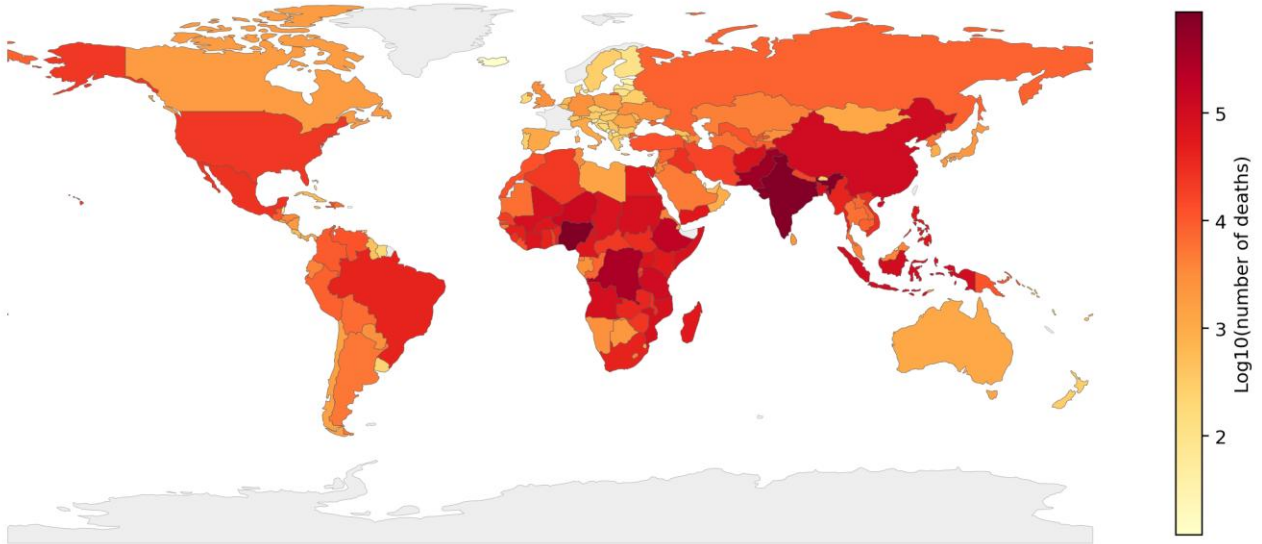
A small preview of the 2020 analysis data is shown below.

Geo	Country	Total child deaths, 2020	Pneumonia child deaths, 2020
afg	Afghanistan	79,117	7,465
ago	Angola	91,357	14,500
alb	Albania	281	10
are	UAE	621	19
arg	Argentina	5,037	179
arm	Armenia	407	32
atg	Antigua and Barbuda	7	0
aus	Australia	1,130	35

World maps for 2020

The maps show country-level differences in each indicator. Because country death counts vary widely, the map colours use a log scale to make differences more visible across countries.

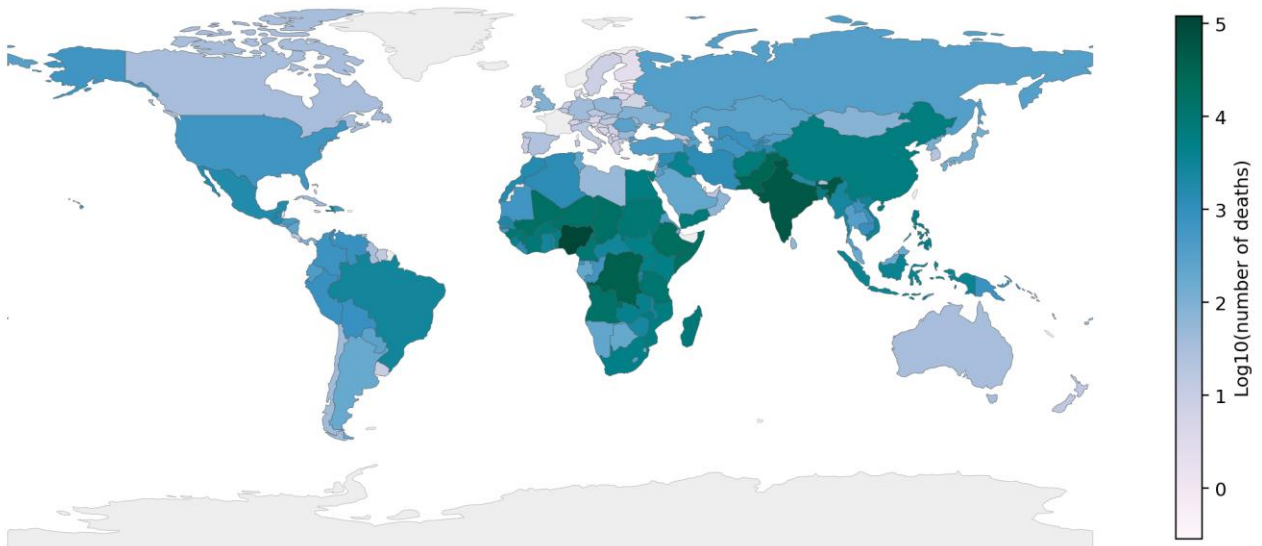
Total child deaths by country, 2020



Countries in grey have no matched 2020 value. Values are shown on a log scale for readability.

Figure 1. Total child deaths by country, 2020.

Pneumonia deaths in children aged 1-59 months by country, 2020



Countries in grey have no matched 2020 value. Values are shown on a log scale for readability.

Figure 2. Deaths due to pneumonia in children aged 1-59 months by country, 2020.

Relationship between the two indicators

The scatterplot compares the two indicators for 2020. The axes use log scales because the number of deaths varies greatly between countries. Points are coloured by pneumonia death level so that lower, medium, and higher pneumonia-death countries are clearly distinguishable.

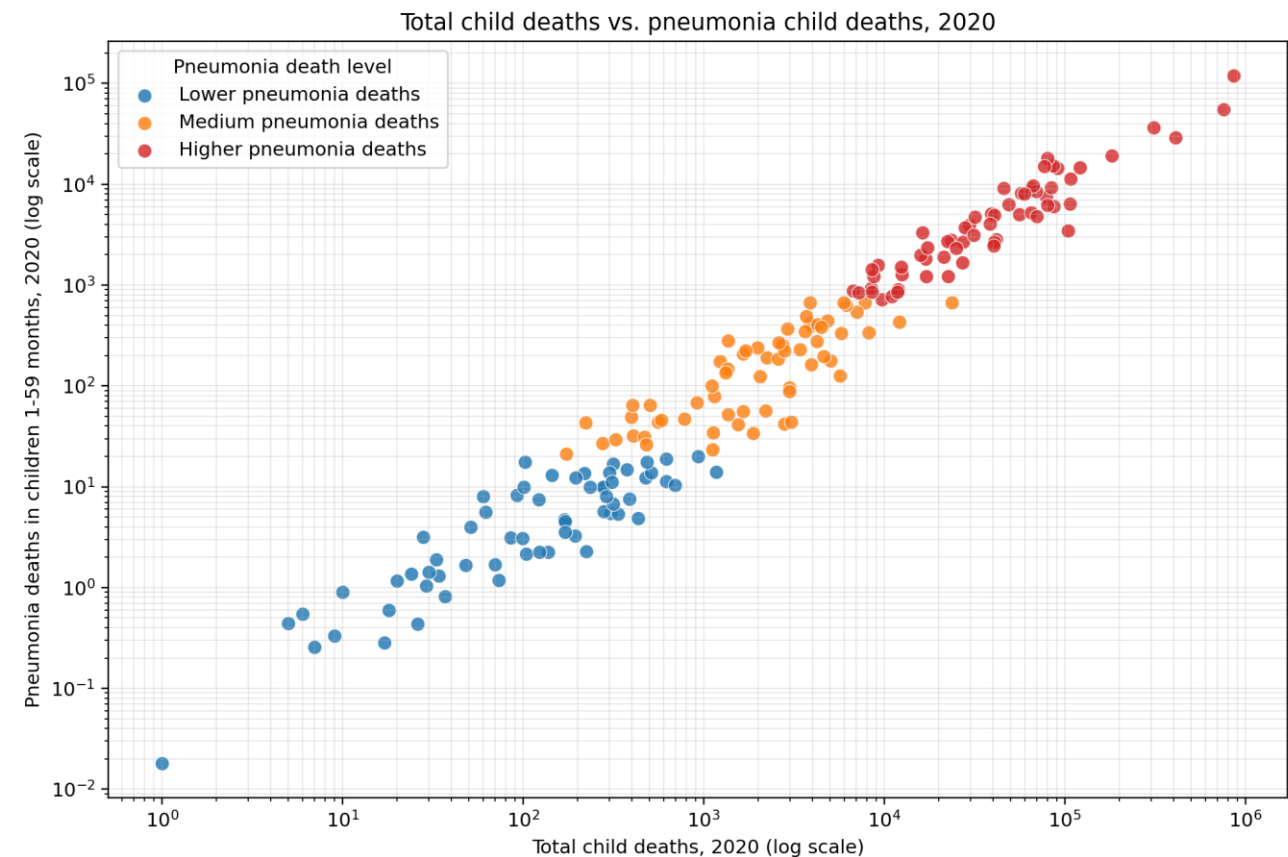


Figure 3. Relationship between total child deaths and pneumonia deaths in children aged 1-59 months, 2020.

Statistic	Value
Countries included in scatterplot	187
Pearson correlation using raw counts	0.950
Spearman correlation using raw counts	0.977
Total child deaths across included countries	5,182,801
Pneumonia child deaths across included countries	552,606

Brief interpretation

The scatterplot shows a strong positive relationship between the total number of child deaths and the number of pneumonia deaths among children aged 1-59 months in 2020. Countries with higher total child deaths generally also have higher numbers of pneumonia-related child deaths.

This pattern is expected because pneumonia deaths form part of overall child mortality. Countries with larger child mortality burdens often also face higher burdens from specific infectious causes, including pneumonia.

The relationship is strong, with a Pearson correlation of 0.950 and a Spearman correlation of 0.977. However, the points are not perfectly aligned, suggesting that pneumonia contributes differently to child mortality across countries. These differences may reflect variation in prevention, vaccination coverage, nutrition, access to care, and health-system conditions.

Reflections:

I found this assignment quite easy with AI. I simply followed the instructions and didn't ask much from it, I just modified the colours. It seems quick and easy. The only challenge I feel is that I need to buy a paid version to get this type of analysis done. The surprising part was the creation of maps and reports in just a few minutes, rather than spending a long time on analysis and report preparation. I also learned how to download data files and use them with AI, and how to give accurate commands for analysis and reporting.